HDOC 5

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Question

Multiply two integers without using the multiplication (*) operator.

The given algorithm and flowchart use repeated addition, but the logic is incorrect/incomplete for some inputs. Study the given algorithm and flowchart, identify the errors, and write a corrected solution that works for positive numbers, negative numbers, and zero while using the minimum possible number of additions.

Given Algorithm:

1. Define num1, num2, prod, index as integers.

2. Get num1, num2.

3. Set prod = 0, index = 1.

4. Repeat steps 4.1 and 4.2 while index <= num2:

4.1 prod = prod + num1

4.2 index = index + 1

5. Print prod.

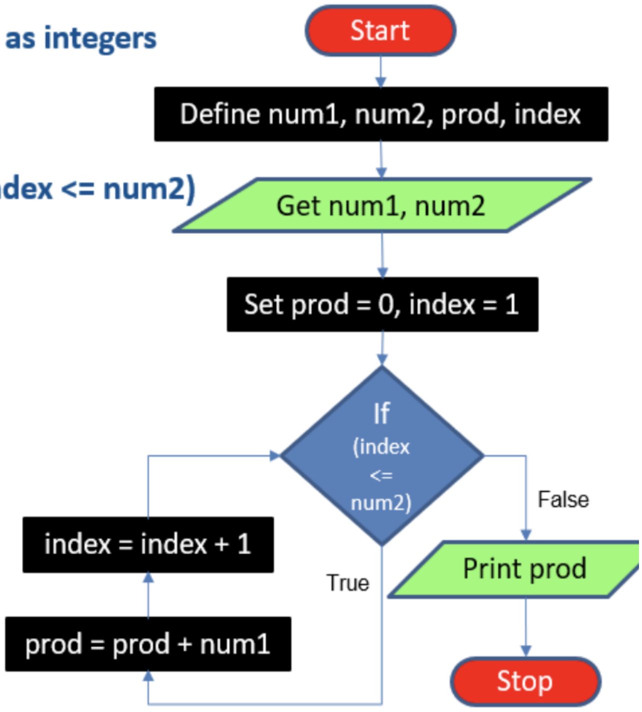
6. Stop.

Given Algorithm & Flowchart (Reference)

Multiply 2 numbers without using *operator (9)

- ## Algorithm & Flowchart

1. Define num1, num2, prod, index as integers
2. Get num1, num2
3. Set prod = 0, index = 1
4. Repeat steps 4.1 and 4.2 until (index <= num2)
   4.1. prod = prod + num1
   4.2. index = index + 1
5. Print prod
6. Stop



Mistakes in the Given Algorithm / Flowchart

1. The loop directly uses num2 as the repetition count, so negative values of num2 are not handled correctly.

2. It does not explicitly handle all sign combinations: positive x negative, negative x positive, and negative x negative.

3. It does not choose the smaller absolute number as the repetition count, so it may perform more additions than necessary.

4. Multiplication by zero is not handled as a separate case.

5. The sign of the final product is not determined from the signs of both input numbers.

6. The corrected flow must use absolute values for repeated addition and apply the sign only after the magnitude is calculated.

Corrected Algorithm

1. Read integers num1 and num2.

2. If either number is 0, set product = 0 and additions = 0; print the result and stop.

3. Find absolute values a = |num1| and b = |num2|.

4. Let count be the smaller of a and b, and addValue be the larger. This gives the minimum number of additions.

5. Set product = 0.

6. Repeat count times: product = product + addValue.

7. If exactly one input is negative, change product to -product.

8. Print product and minimum additions (= count).

9. Stop.

C Program

```c
#include <stdio.h>

int main() {
    int num1, num2, a, b;
    int product = 0;
    int count, addValue, i;

    printf("Enter two integers: ");
    scanf("%d %d", &num1, &num2);

    if (num1 == 0 || num2 == 0) {
        printf("Product: 0\n");
        printf("Minimum additions: 0\n");
        return 0;
    }

    a = (num1 < 0) ? -num1 : num1;
    b = (num2 < 0) ? -num2 : num2;

    if (a < b) {
        count = a;
        addValue = b;
    } else {
        count = b;
        addValue = a;
    }

    for (i = 0; i < count; i++)
        product = product + addValue;

    if ((num1 < 0 && num2 > 0) ||
        (num1 > 0 && num2 < 0))
        product = -product;

    printf("Product: %d\n", product);
    printf("Minimum additions: %d\n", count);

    return 0;
}
```

Output / Verification

Input	Product	Minimum Additions
12 3	36	3
3 12	36	3
-6 5	-30	5
6 -5	-30	5
-9 -4	36	4
-4 -9	36	4
0 8	0	0
8 0	0	0
0 -7	0	0
-7 0	0	0
0 0	0	0
1 -7	-7	1
-1 -9	9	1

Example Console Output

```
Enter two integers: 12 3
Product: 36
Minimum additions: 3

Enter two integers: 6 -5
Product: -30
Minimum additions: 5

Enter two integers: -9 -4
Product: 36
Minimum additions: 4

Enter two integers: 0 8
Product: 0
Minimum additions: 0
```